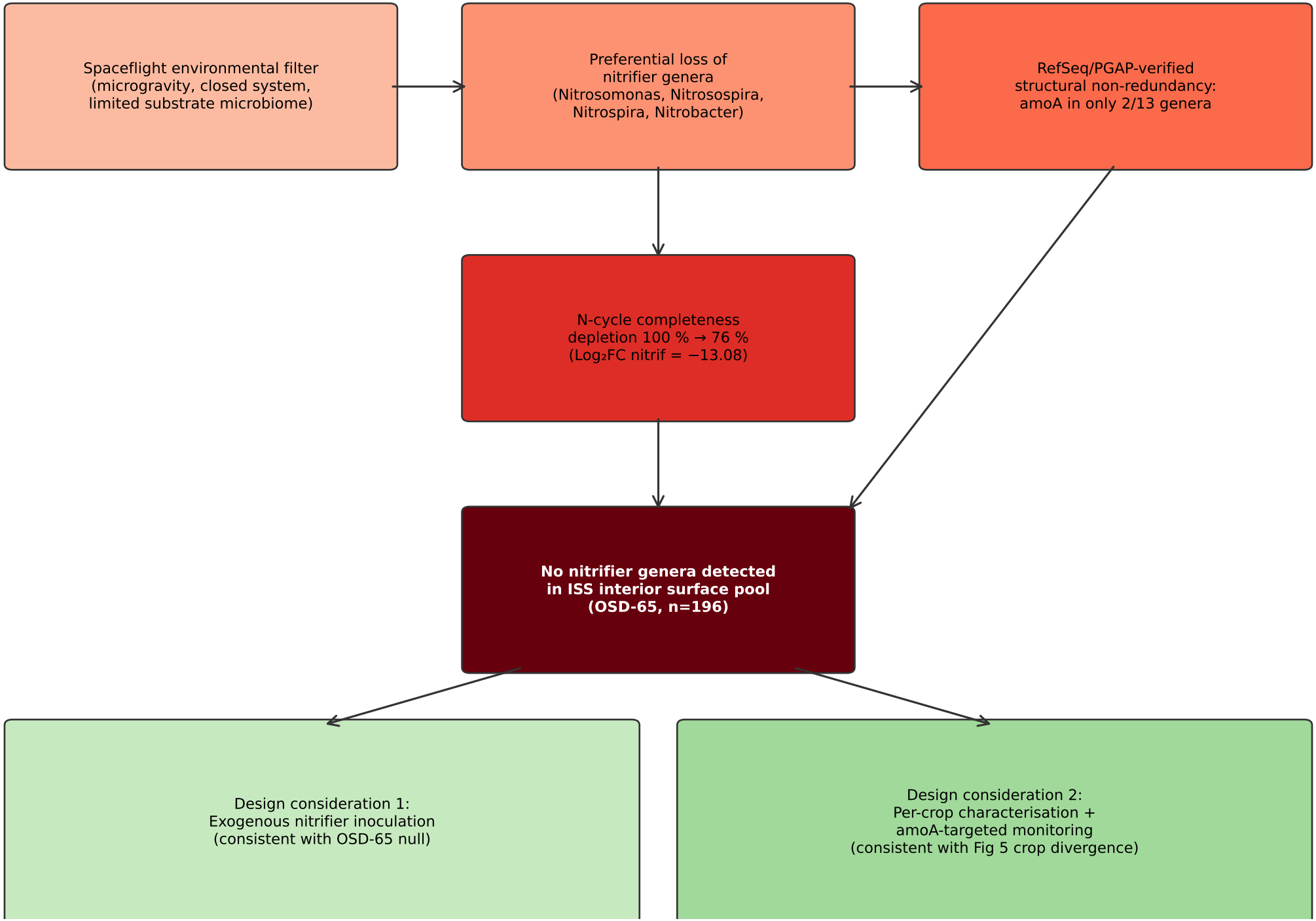
Spaceflight environmental filter
(microgravity, closed system,
limited substrate microbiome)
Preferential loss of
nitrifier genera
(Nitrosomonas, Nitrosospira,
Nitrospira, Nitrobacter)
RefSeq/PGAP-verified
structural non-redundancy:
amoA in only 2/13 genera
N-cycle completeness
depletion 100 % → 76 %
(Log₂FC nitrif = −13.08)
No nitrifier genera detected
in ISS interior surface pool
(OSD-65, n=196)
Design consideration 1:
Exogenous nitrifier inoculation
(consistent with OSD-65 null)
Design consideration 2:
Per-crop characterisation +
amoA-targeted monitoring
(consistent with Fig 5 crop divergence)